

password hidden in the executable file *test1*. The command *strings* prints the strings of printable characters in a file. So it can be used to print the human-readable strings in an executable file. The command *strings* is a part of the GNU Binary Utilities (Binutils). Figure 2 shows a part of the output when the command *strings test1* is executed.

From Figure 2, we can clearly see that trying out the word *hello* is a very logical move. Figure 3 shows the output of the program *crack1.c* when provided with the password *hello*.

What if we modify the program *crack1.c* by replacing the line of code:

```
#define pass "hello"
```

...with the line of code:

```
#define pass "123456"
```

On execution of the command *strings test1* we can get the password as *123456* without much difficulty because the number *123456* is also a human readable string. Now let us use the Linux command *file* to obtain information about the executable file *test1*. The command *file* is used to determine the file type.

Figure 4 shows the output of the command *file test1*. From Figure 4, along with other information, we can also see that the executable file *test1* is not a stripped file — one in which debugging information and other data not required for program execution is removed so that the resulting executable file is of a smaller size. Let us strip the executable file *test1* with the command *strip* hoping that stripping will hide some information from a potential reverse engineering attack. Figure 4 also shows the execution of the *strip* command, followed by the *file* command which shows the changes made by the *strip* command. Using the command *ls*, the size of the file before and after the strip operation is also shown in Figure 4.

```
File Edit View Search Terminal Help
# gcc crack1.c -o test1
# ./test1

Enter the password: abcd

Sorry
```

Figure 1: An incorrect password on *crack1.c*

```
File Edit View Search Terminal Help
=9
AWAVI
AUATL
[]A\A]A^A_
Enter the password:
hello
You have CRACKED me!!!
Sorry
;*3$"
GCC: (Ubuntu 7.3.0-27ubuntu1~18.04) 7.3.0
crtstuff.c
```

Figure 2: Output of the *strings* command on *test1*

```
File Edit View Search Terminal Help
# ./test1

Enter the password: hello

You have CRACKED me!!!
```

Figure 3: The correct password on *crack1.c*

From Figure 4, we can see that the file size reduces from 8488 bytes to 6120 bytes after the strip operation. Let us now use the command *strings* again to check whether the stripped executable file *test1* hides the password or not. On execution of the command

strings test1, we can see that the password is not hidden in the stripped executable file *test1* and the output is the same as the one shown in Figure 2. So, stripping the executable file will not complicate the reverse engineering process much.

Now, let us use a utility called *objdump* which shows the information from an object file. Figure 5 shows a part of the output from the execution of the command *objdump -s test1*. The utility *objdump* is also a part of the GNU Binary Utilities (Binutils). It can be used as a disassembler also. But we will only use the option *-s* which is used to show the full contents of all the sections.

From Figure 5, we can easily guess that the password is *hello*. Another GNU Binary Utility called *readelf* is also very useful and it is similar in performance to *objdump*. Though *strings* and *objdump* are relatively powerful utilities, we can easily hide the password from exploitation with *strings* and *objdump* utilities. Simple hacks can be used to make reverse engineering more difficult. For example, consider the modified C program called *crack2.c* shown below.

```
File Edit View Search Terminal Help
# ls -l test1
-rwxr-xr-x 1 deepu deepu 8488 Feb 11 02:29 test1
# file test1
test1: ELF 64-bit LSB shared object, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, for GNU/Linux 3.2.0, BuildID[sha1]=e ca1cc1d33644b5553d9229ead1b9aa7962a24cb, not stripped
# strip test1
# file test1
test1: ELF 64-bit LSB shared object, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, for GNU/Linux 3.2.0, BuildID[sha1]=e ca1cc1d33644b5553d9229ead1b9aa7962a24cb, stripped
# ls -l test1
-rwxr-xr-x 1 deepu deepu 6120 Feb 11 02:30 test1
```

Figure 4: Output of the *file*, *strip* and *ls* commands

```
File Edit View Search Terminal Help
Contents of section .rodata:
08c0 01000200 0a456e74 65722074 68652070 .....Enter the p
08d0 61737377 6f72643a 20002573 0068656c assword: %.s.hel
08e0 6c6f000a 596f7520 68617665 20435241 lo..You have CRA
08f0 434b4544 206d6521 2121000a 536f7272 CKED me!!!..Sorr
0900 7900 y.
Contents of section .eh_frame_hdr:
0904 011b033b 38000000 06000000 2cfdffff ...;8.....
0914 84000000 8cfdffff ac000000 9cfdffff .....
0924 54000000 a6feffff c4000000 3cfdffff T.....<...
0934 e4000000 acfdffff 2c010000 .....
```

Figure 5: Output of the *objdump* utility on *test1*